

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25-06-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	User Registration, Secure Login, Password Authentication
FR-2	User Confirmation	Accept User Input, Text Preprocessing, Emotion Prediction using BiLSTM & BERT, Confidence Score Generation
FR-3	AI-Powered Learning Support	Gemini AI Integration, Personalized Learning Suggestions, Motivational Responses, Emotion-Based Guidance
FR-4	Dashboard & Analytics	Display Emotion Results, Model Comparison, Emotion History, CSV Logging, Analytics Dashboard

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide an intuitive and user-friendly Streamlit interface for students and educators.
NFR-2	Security	Protect user credentials, API keys, and emotion records using secure authentication and encrypted communication.
NFR-3	Reliability	Deliver consistent emotion predictions and AI-generated responses with minimal failures.
NFR-4	Performance	Generate emotion predictions and personalized responses within a few seconds.
NFR-5	Availability	Ensure the application is available whenever users need academic assistance through cloud deployment

NFR-6	Scalability	Support multiple users simultaneously and allow future expansion with additional AI models and educational features.
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